A Comparative Study for Brain Tumor Segmentation in MRI Scans

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Executive Summary

Accurate brain tumor segmentation from MRI is a fundamental but challenging problem in medical computer vision. Unlike image classification, which predicts a single label for an image, segmentation requires precise pixel-level delineation of pathological tissue. This substantially increases task difficulty due to severe class imbalance, ambiguous boundaries, and sensitivity to small localization errors. In clinical contexts, segmentation quality directly affects downstream interpretation, treatment planning, and quantitative monitoring, making reliability as important as raw benchmark accuracy.

This project investigates deep learning methods for binary brain tumor MRI segmentation using U-Net-based architectures, but extends beyond a conventional architecture benchmark. The central goal is not merely to identify the highest-performing model, but to examine how segmentation conclusions change under different evaluation protocols, failure conditions, uncertainty analysis, and cross-dataset transfer scenarios.

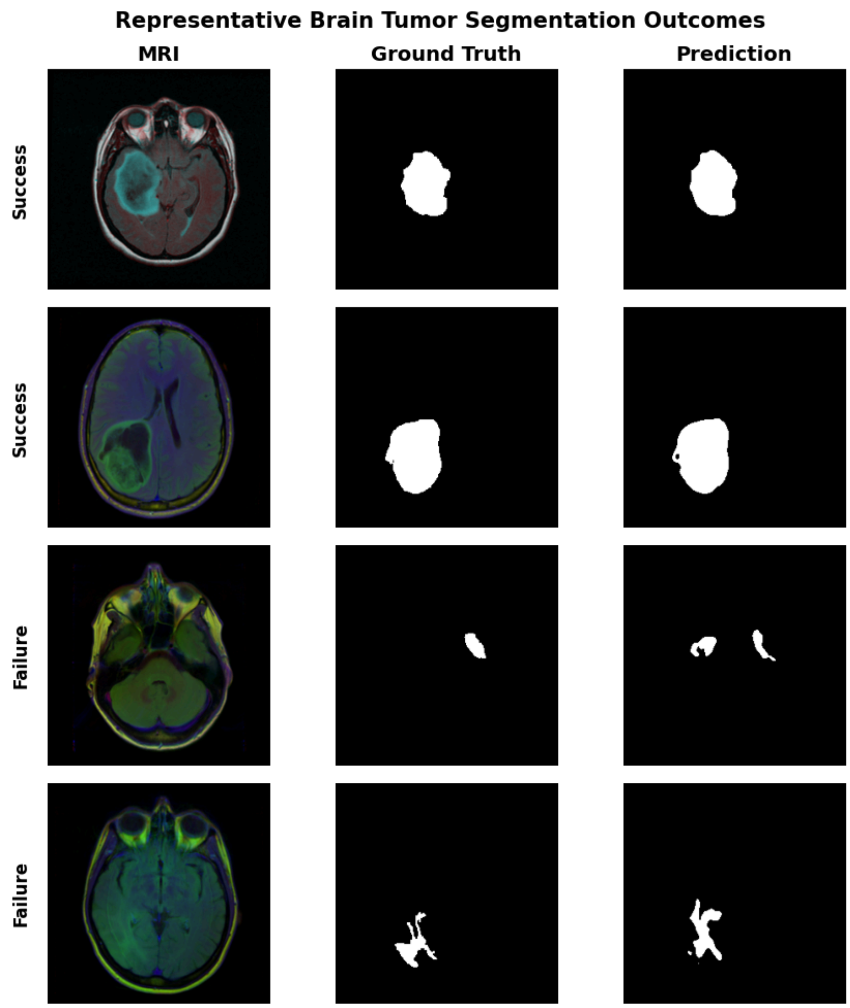
Four segmentation architectures were evaluated: a Vanilla U-Net implemented from scratch, Attention U-Net, ResNet34 U-Net, and EfficientNet-B0 U-Net. Additional experiments explored loss-function design, augmentation strategy, lesion-size effects, failure categorization, inference efficiency, prediction confidence calibration, robustness under image corruption, and transfer learning to an external MRI dataset.

Under standard full-dataset evaluation, EfficientNet-B0 U-Net achieved the strongest Dice score (0.9117), substantially outperforming simpler baselines. However, this apparent architectural advantage weakened when evaluation was restricted to tumor-containing slices only, where EfficientNet-B0 and ResNet34 achieved nearly identical performance (0.8538 vs. 0.8493 Dice).

Further analysis identified several clinically relevant limitations. Performance degraded sharply on very small tumors (0.674 Dice).Finally, transfer learning experiments showed poor zero-shot generalization to an external dataset (0.452 Dice) but strong recovery after fine-tuning (0.816 Dice), suggesting that learned segmentation features remain transferable despite substantial domain shift.

Overall, the project evolved from a model comparison into a broader investigation of segmentation evaluation methodology, reliability, and generalization in medical AI.

Figure 1. Representative successful and failure segmentation cases from the best-performing model.



1.Introduction

Medical image segmentation is one of the most technically demanding tasks in computer vision because it requires dense structured prediction at the pixel level rather than global image-level categorization. In medical contexts, this difficulty becomes especially important because prediction errors are not merely academic benchmark artifacts; they may directly affect clinical interpretation.

Brain tumor segmentation is a particularly relevant application. Accurate delineation of tumor boundaries can support diagnosis. However, the task remains challenging because tumors vary substantially in size, morphology, anatomical location, contrast, and visual appearance. Some lesions are large and visually distinct, while others are small, irregular, low-contrast, or partially ambiguous even to human observers.

A further challenge is severe class imbalance. In MRI segmentation, most pixels belong to healthy tissue or background, while tumor pixels often represent only a small minority of the image. This imbalance can distort optimization and produce misleading benchmark interpretations if evaluation protocols are not carefully defined.

Deep learning has become the dominant paradigm for medical image segmentation, with U-Net serving as the Standard Segmentation Framework. U-Net introduced the encoder-decoder design with skip connections, enabling simultaneous high-level semantic reasoning and preservation of spatial detail. Since its introduction, numerous variants have been proposed, including residual encoder hybrids, attention-augmented architectures, and computationally optimized backbones.

However, simple benchmark comparison is often insufficient. High average Dice scores may conceal clinically meaningful failure modes, dependence on evaluation protocol, poor robustness, or lack of generalization across datasets.

This project goes beyond a simple architecture comparison and examines several broader aspects of segmentation reliability, including evaluation methodology, lesion-size sensitivity, prediction confidence, robustness under image corruption, and cross-dataset transfer learning. The goal is not only to identify a strong-performing model, but also to better understand when segmentation systems succeed, when they fail, and how reliable their predictions are under different conditions.

Rather than focusing narrowly on maximizing a single benchmark score, this project aims to evaluate segmentation quality from a broader reliability perspective.

2. Method

2.1 Datasets

The primary dataset used in this project was a brain MRI segmentation dataset consisting of two-dimensional MRI slices paired with binary tumor masks. Each sample includes an MRI image and a corresponding ground-truth segmentation mask indicating tumor regions. A notable characteristic of this dataset is that many slices contain no tumor pixels at all. These are valid clinical images, but they create an important evaluation issue because predicting an empty mask in these cases is trivially correct and can artificially inflate segmentation metrics.

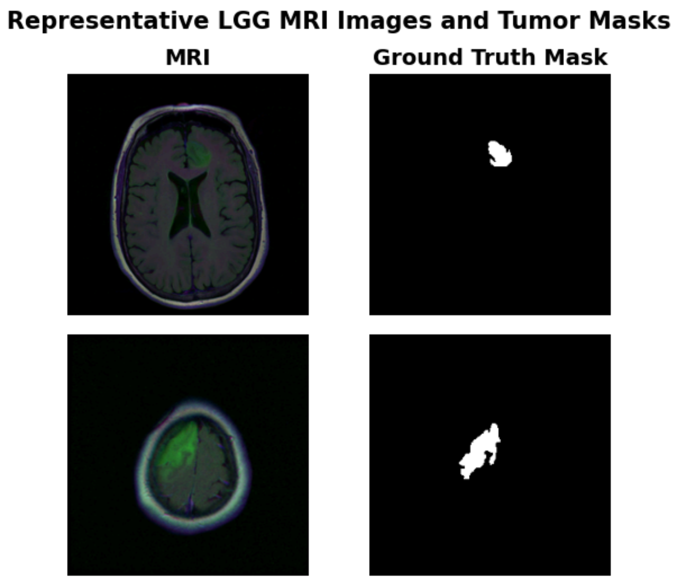


Figure 2. Representative MRI slices from the LGG dataset and their corresponding ground-truth tumor masks.

To address this, two evaluation protocols were defined. The first used the full dataset, including all slices regardless of whether tumor pixels were present. This reflects the natural data distribution but partially rewards correct background prediction. The second protocol used only tumor-containing slices, creating a stricter benchmark that evaluates actual segmentation quality when pathology is present. Comparing these two protocols became an important part of the analysis.

To evaluate generalization, an external dataset (BRISC2025) was also introduced for transfer learning experiments. This dataset was not used for architecture benchmarking, but rather to examine whether models trained on the primary dataset could generalize across a different imaging distribution.

2.2 Preprocessing and Augmentation

All images were resized to 256 × 256 pixels before training. MRI images were normalized, and segmentation masks were converted into binary format. A custom PyTorch dataset pipeline was implemented to handle image loading, mask processing, normalization, and tensor conversion in a consistent way across all experiments.

To study regularization effects, three augmentation settings were tested. The no-augmentation setting used only resizing, normalization, and tensor conversion. The light augmentation setting added horizontal and vertical flips, small rotations, and brightness changes. The heavy augmentation setting used stronger flipping and rotation probabilities, shift-scale-rotate transformations, Gaussian noise, and Gaussian blur. These experiments were designed to test whether stronger augmentation improves generalization or instead hurts performance by perturbing medically meaningful image structure.

2.3 Model Architectures

Four U-Net-based architectures were evaluated. The first was a Vanilla U-Net implemented from scratch, which served as the simplest baseline. This model does not use pretrained encoders or additional architectural modifications, making it a useful reference point.

Attention U-Net was included to test whether spatial attention improves segmentation quality by encouraging the model to focus more strongly on relevant image regions. While attention mechanisms are often attractive in medical imaging, their practical benefit depends on whether the underlying feature representations are already strong.

Two pretrained encoder architectures were also evaluated. ResNet34 U-Net replaces the standard encoder with a ResNet34 backbone initialized from pretrained weights, providing stronger hierarchical feature extraction and more stable optimization. EfficientNet-B0 U-Net similarly uses a pretrained encoder, but with a more parameter-efficient backbone designed for strong performance at lower computational cost.

2.4 Loss Functions

Medical image segmentation suffers from severe class imbalance because tumor pixels occupy only a small portion of the image. To study how optimization objectives affect segmentation behavior, three loss configurations were evaluated: BCE-Dice, Dice-only, and Tversky loss.

The primary configuration combined Binary Cross-Entropy and Dice loss, balancing pixel-level supervision with direct overlap optimization. This hybrid objective provided stable optimization while encouraging accurate tumor region prediction.

Tversky loss was included as an alternative designed to place greater emphasis on false-negative reduction, which is particularly relevant for small tumor detection. Dice-only loss was also evaluated as a simpler overlap-focused baseline.Among these options, BCE-Dice consistently produced the strongest overall performance and was therefore used in the main architecture comparisons.

2.5 Evaluation Design

Performance was evaluated using Dice score, Intersection over Union (IoU), precision, and recall, with Dice treated as the primary metric.

Beyond standard benchmarking, several additional analyses were conducted. Tumor size analysis examined whether segmentation quality changes across lesions of different sizes. Failure analysis manually categorized poor predictions into clinically interpretable groups such as missed tumors, false positives, and partial boundary failures.

Confidence analysis was also performed to test whether failed predictions showed lower confidence than successful ones. Inference benchmarking compared model efficiency using parameter count and runtime per image. Finally, transfer learning experiments evaluated both zero-shot and fine-tuned performance on BRISC2025.

3. Results

3.1 Main Architecture Comparison

The primary benchmark used the positive-only evaluation protocol because it focuses directly on tumor segmentation rather than rewarding correct background prediction. Under this stricter setting, EfficientNet-B0 U-Net achieved the strongest Dice score of 0.8538, followed closely by ResNet34 U-Net at 0.8493. The gap between these two models was small, suggesting that both pretrained encoder architectures perform similarly when evaluation focuses only on tumor-containing slices.

Vanilla U-Net performed substantially worse, achieving a Dice score of 0.7855. This indicates that pretrained encoder quality contributes significantly to segmentation performance. Attention U-Net improved over the baseline but remained below the pretrained encoder models, reaching 0.8116 Dice. This suggests that adding architectural complexity through attention does not necessarily outperform simpler models with stronger feature extractors.

Table 1. Main architecture comparison under different evaluation protocols

Model	Full Dice	Positive-only Dice	IoU	Precision	Recall
Vanilla U-Net	0.834	0.785	0.658	0.711	0.905
ResNet34 U-Net	0.857	0.849	0.766	0.852	0.879
EfficientNet-B0 U-Net	0.912	0.854	0.770	0.850	0.882

3.2 Evaluation Protocol Affects Benchmark Conclusions

When the full dataset was used instead of the positive-only subset, the benchmark results changed significantly. Under full evaluation, EfficientNet-B0 U-Net achieved a Dice score of 0.9117, substantially higher than ResNet34 U-Net at 0.8566. At first glance, this suggests a clear architectural advantage.

However, this interpretation becomes less convincing when compared with the positive-only benchmark, where the difference between EfficientNet and ResNet34 was much smaller. The most likely explanation is that full-dataset evaluation includes many slices with empty masks, where predicting background everywhere is correct. This increases segmentation metrics without necessarily reflecting meaningful tumor localization performance.

This result became one of the most important findings in the project. The apparent superiority of a model can depend strongly on evaluation design, especially in medical segmentation datasets where negative slices are common. Reporting only the highest Dice score without considering this effect would give an incomplete picture of model quality.

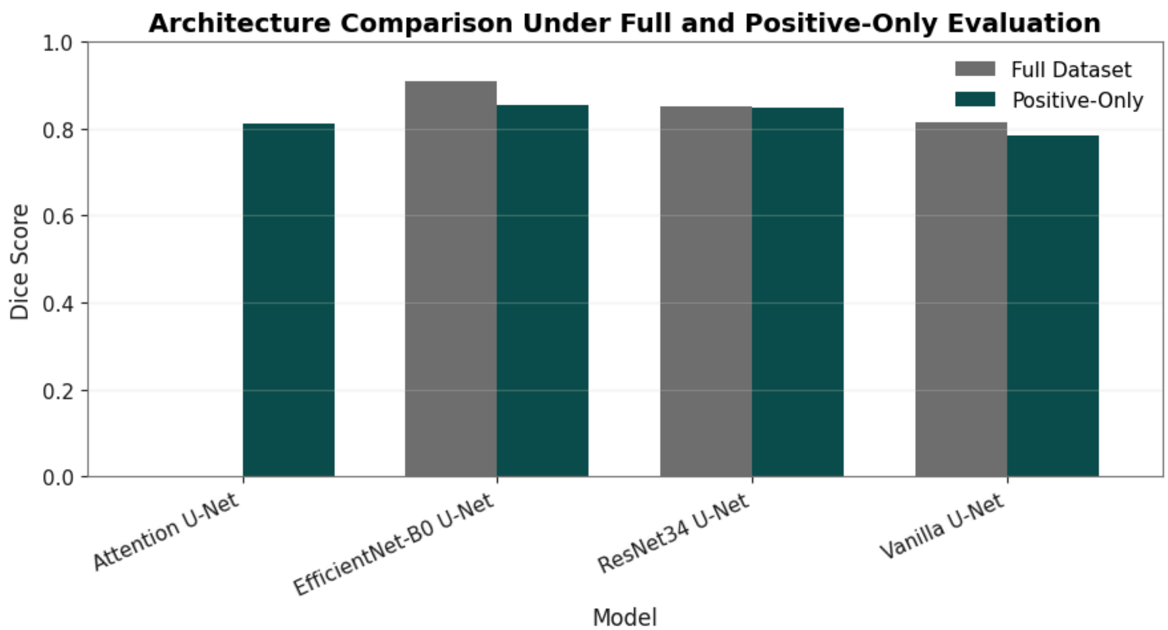


Figure 3. Full-dataset vs positive-only benchmark comparison.

3.3 Loss Function Comparison

Different loss functions produced noticeably different segmentation behavior. Among the tested losses, the BCE-Dice combination achieved the strongest overall performance, with ResNet34 U-Net reaching a Dice score of 0.8493 under positive-only evaluation.

Dice-only loss produced slightly weaker overlap performance at 0.8203 Dice, while Tversky loss achieved 0.8233. Although Tversky did not produce the best overall Dice score, it achieved the highest recall, suggesting that it encouraged more aggressive tumor prediction and reduced missed tumor regions at the cost of more false positives.

This tradeoff is meaningful in medical segmentation, where false negatives may be more concerning than over-segmentation in some applications. However, for overall balanced performance, the BCE-Dice hybrid consistently performed best and was therefore used as the primary loss configuration.

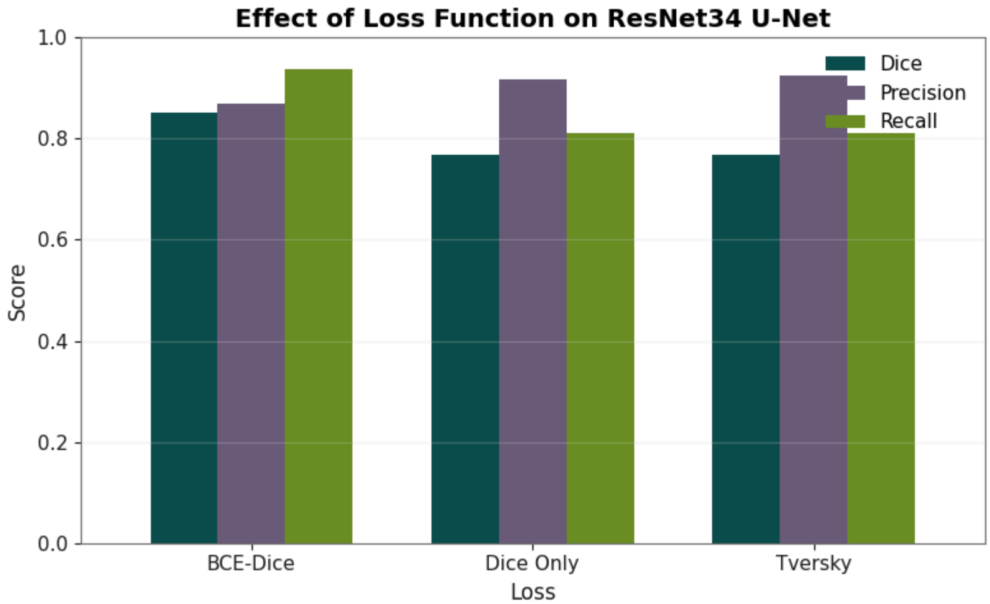


Figure 4. Loss function comparison.

3.4 Tumor Size Analysis

Tumor size had a major effect on segmentation quality. Medium tumors achieved the strongest performance, with an average Dice score of 0.9227. Small tumors remained relatively strong at 0.8937 Dice, but very small tumors showed a substantial performance drop to 0.6738.

This gap is important. A model that appears strong based on average benchmark performance may still struggle with precisely the cases that are hardest to detect. Very small tumors occupy only a limited number of pixels, making them highly sensitive to localization error and boundary uncertainty.

The increased variance for very small lesions also suggests inconsistent behavior across cases. Some were segmented well, while others were missed or only partially detected. This analysis shows why aggregate benchmark metrics should be interpreted carefully.

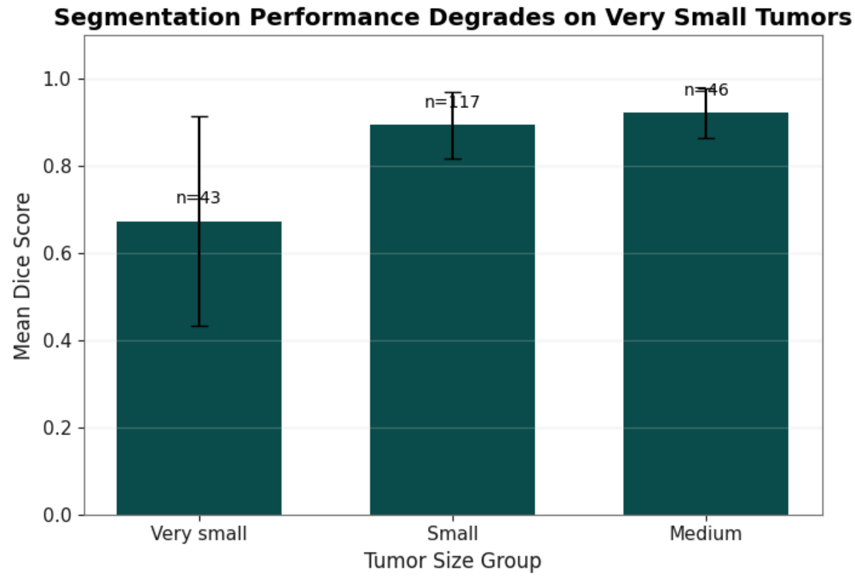


Figure 5. Performance by tumor size.

3.5 Failure Analysis

Under positive-only evaluation, most cases were categorized as good segmentations, but several failure patterns appeared repeatedly.

The most concerning category was missed tumors, where the model failed to identify the lesion entirely. These cases were often associated with very small tumors or low-contrast boundaries. Partial boundary failures also appeared, where the model detected the tumor but produced incomplete or inaccurate boundaries. A smaller number of false-positive-heavy cases showed segmentation in anatomically incorrect regions.

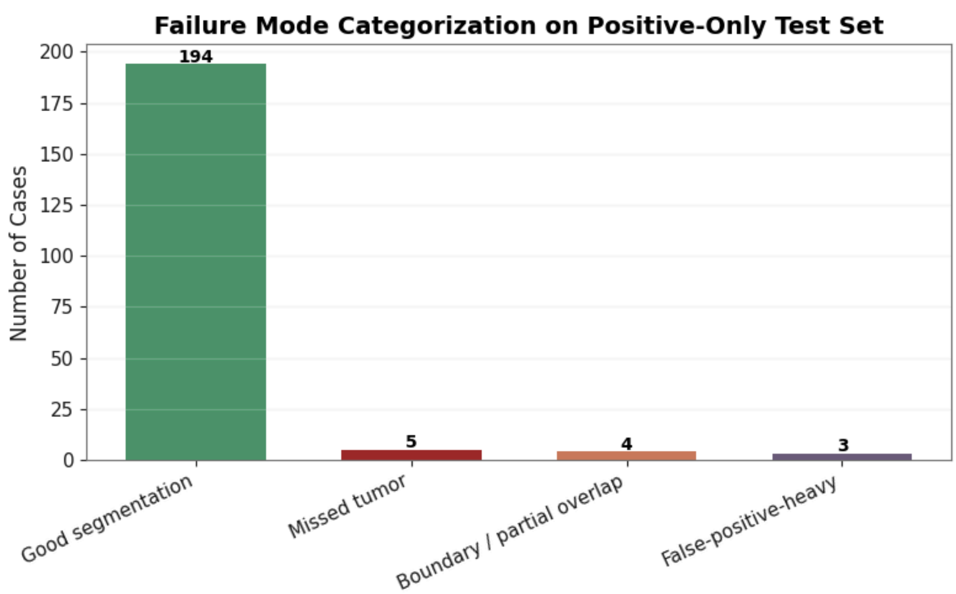


Figure 6. Representative failure cases.

The full-dataset analysis showed similar patterns but with a larger total number of failures due to the larger test set.

This analysis was important because average metrics alone cannot distinguish between clinically different failure types. A slightly inaccurate boundary and a completely missed tumor may produce similar quantitative penalties, but their practical significance is very different.

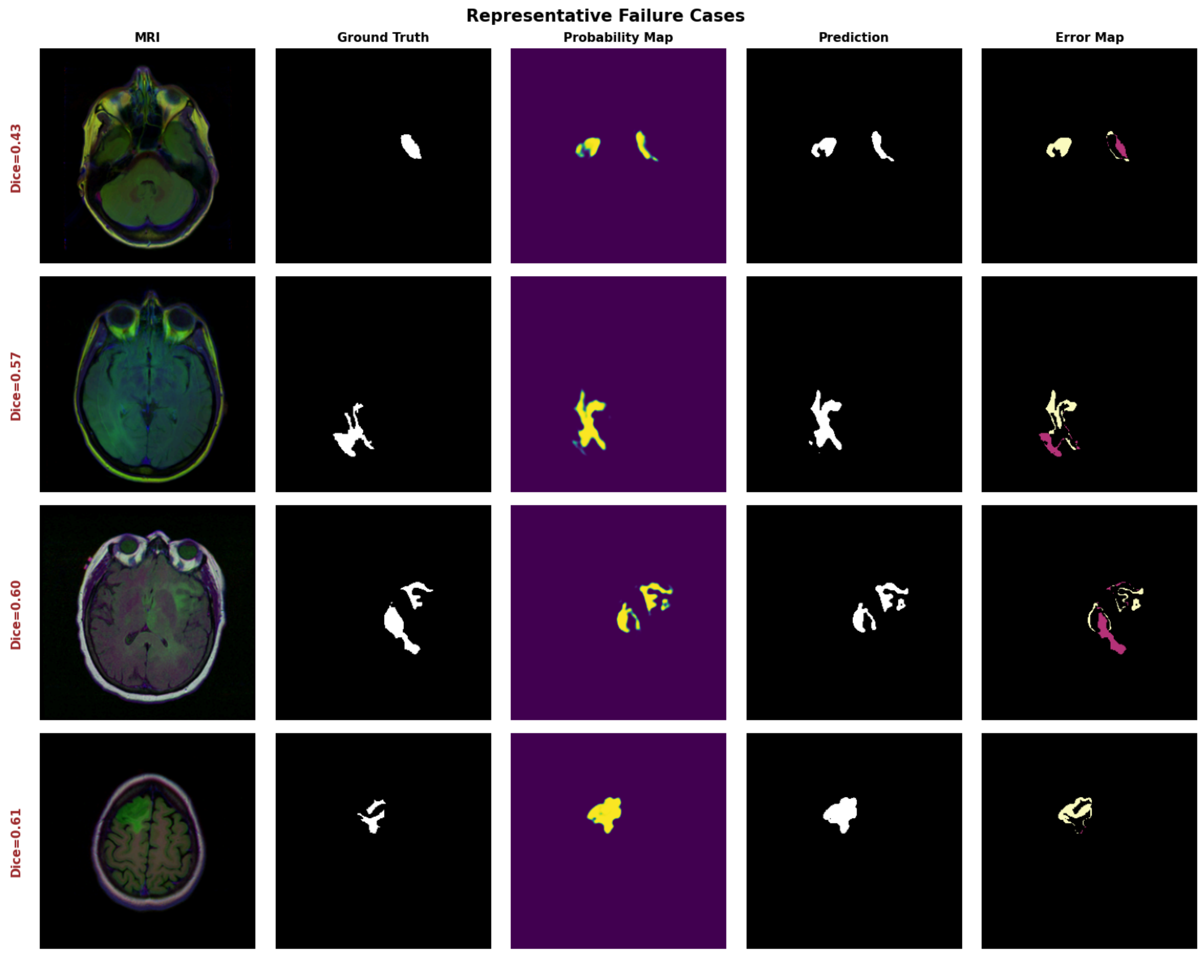


Figure 7. Representative failure cases from the best-performing model. Examples include missed boundaries, false positives, and partial overlap errors. Probability maps and error visualizations reveal that some incorrect predictions were still produced with high confidence.

3.6 Efficiency Comparison

Model efficiency was evaluated using both parameter count and inference speed. EfficientNet-B0 achieved the strongest segmentation performance while using the fewest parameters among the pretrained models, at approximately 6.25 million parameters. ResNet34 used roughly 24.4 million parameters, while Vanilla U-Net contained approximately 7.8 million.

Despite these differences, inference speed remained nearly identical, with all models requiring roughly 15 milliseconds per image.

This suggests that EfficientNet's performance advantage was not achieved by simply increasing computational cost. From a practical perspective, EfficientNet offered the strongest balance between segmentation quality and efficiency.

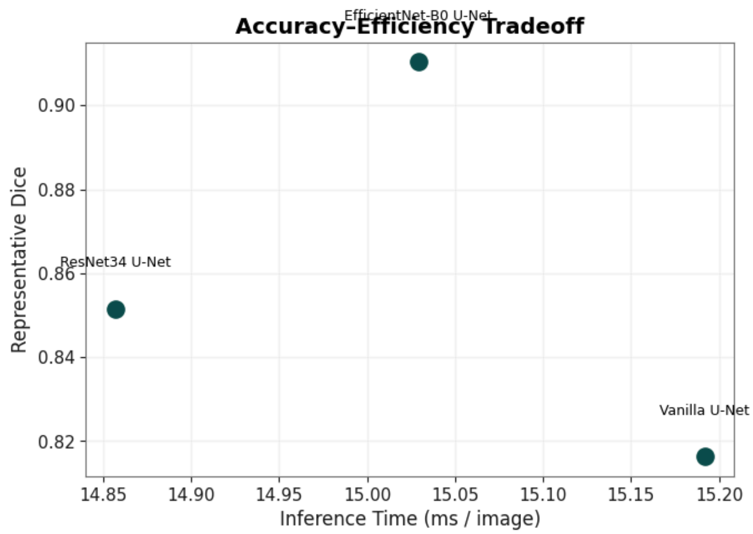


Figure 8. Accuracy-efficiency comparison.

3.7 Confidence and Reliability Analysis

One of the most interesting findings came from the confidence analysis. A reasonable expectation is that failed segmentations should show lower confidence than successful ones. However, the results suggested the opposite.

Failed cases had an average Dice score of 0.3738, but their average confidence remained extremely high at 0.9977. High-quality segmentations achieved a much stronger Dice score of 0.9240, yet their average confidence was slightly lower at 0.9964.

This suggests that the model's confidence estimates are poorly calibrated. In other words, the model often appears highly certain even when the segmentation is clearly incorrect. This is an important reliability concern, particularly in medical applications where confidence might otherwise be used as a safety signal.

A high-confidence wrong prediction is more concerning than a low-confidence uncertain one because it may appear trustworthy despite being incorrect. These results suggest that raw prediction confidence should not be treated as a reliable indicator of segmentation quality.

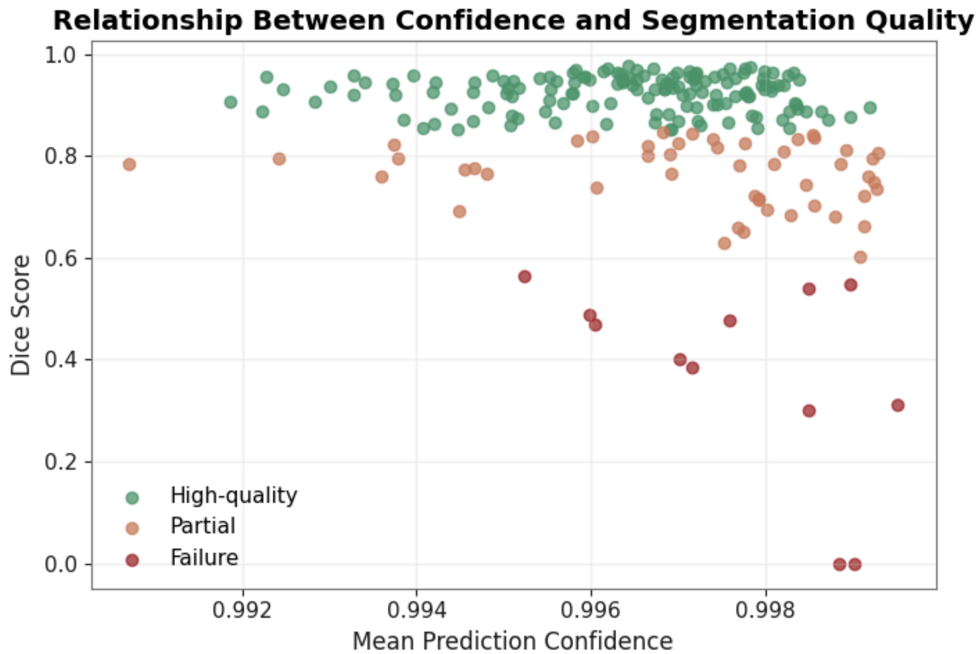


Figure 9. Confidence vs segmentation quality.

3.8 Transfer Learning and Cross-Dataset Generalization

To evaluate generalization beyond the primary dataset, the best-performing in-domain model was tested on BRISC2025.

In the zero-shot setting, where the model was evaluated without retraining, performance dropped sharply. The Dice score fell to 0.4516, with similarly weak IoU, precision, and recall. This indicates substantial domain shift between the original dataset and BRISC.

Several factors could explain this drop, including differences in imaging protocols, scanner characteristics, tumor appearance, or annotation conventions. Strong in-domain performance clearly did not translate into immediate out-of-domain generalization.

However, fine-tuning produced a very different result. After transfer learning, performance improved dramatically to 0.8162 Dice.

This is one of the strongest positive findings in the project. Although zero-shot deployment failed, the pretrained segmentation representations remained highly transferable after adaptation. This suggests that the learned features were useful, but direct generalization across datasets was limited.

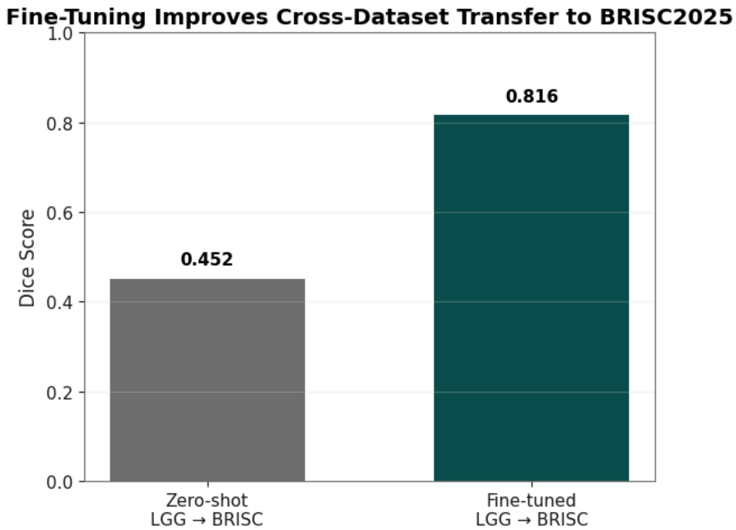


Figure 10. BRISC zero-shot vs fine-tuned transfer performance.

4. Discussion and Conclusion

This project began as a comparison of segmentation architectures, but the final results revealed a broader set of lessons about reliability, evaluation design, and generalization.

The most obvious conclusion is that pretrained encoder architectures clearly outperform a scratch-trained baseline. Both ResNet34 and EfficientNet consistently achieved much stronger segmentation performance than Vanilla U-Net, suggesting that feature quality plays a major role in medical segmentation performance. Attention U-Net improved over the baseline but did not outperform the pretrained encoder models, indicating that adding architectural complexity alone does not guarantee better results.

A more important finding was that evaluation design significantly affects benchmark interpretation. Under full-dataset evaluation, EfficientNet appeared dramatically superior, achieving 0.9117 Dice. However, when evaluation was restricted to tumors containing slices only, the performance gap narrowed substantially. This suggests that some benchmark gains are influenced by the inclusion of empty-mask slices, where predicting background is trivially correct. For medical segmentation tasks, benchmark methodology matters as much as architecture choice.

The tumor size analysis highlighted another important limitation. While average performance was strong, very small tumors remained difficult to segment reliably. This is clinically meaningful because small lesions may be among the most important cases to detect accurately. Aggregate Dice scores can hide this weakness.

The confidence analysis revealed an especially concerning result: failed predictions were often highly confident. This suggests poor calibration and limits the usefulness of raw confidence as a practical reliability signal. A segmentation system that is confidently wrong creates additional deployment risk.

The transfer learning experiments produced a more balanced conclusion. Zero-shot generalization to BRISC was poor, showing that strong performance on one dataset does not imply broad generalization. However, the strong improvement after fine-tuning suggests that learned segmentation features remain useful across domains. This makes transfer learning a promising strategy even when direct deployment is not realistic.

There are several limitations to this project. First, the main benchmarking experiments were based on a single primary dataset, which limits broader generalization claims. Second, the models operated on 2D slices rather than full 3D MRI volumes, meaning spatial context was not used. Third, confidence analysis relied on simple confidence estimates rather than more formal uncertainty calibration methods.

Overall, this project shows that high segmentation accuracy alone is not enough to evaluate medical systems. Failure behavior, robustness, and generalization all matter. The strongest model in a benchmark is not necessarily the most trustworthy model in practice.

5.Statement of Individual Contribution

This project was completed independently. All aspects of the work, including experiment design, implementation, model training, evaluation, visualization, analysis, and report writing, were completed by the author.

6.Statement on AI Use

Tools such as Gemini , ChatGPT were used to help clarify deep learning concepts, including segmentation architectures and loss functions. They were also used to assist with debugging code and suggesting potential improvements to the training pipeline.
In addition, AI tools were used to provide general guidance on experiment design and to help organize ideas for the report. However, all model implementations, experiments, and analyses were performed independently.

7.References

U-Net: Convolutional Networks for Biomedical Image Segmentation

https://arxiv.org/abs/1505.04597

Exploring Vanilla U-Net for Lesion Segmentation from Whole-body FDG-PET/CT Scans

https://arxiv.org/abs/2210.07490

Segmentation Models PyTorch

https://github.com/qubvel/segmentation_models.pytorch

Kaggle Brain MRI Dataset

https://www.kaggle.com/datasets/mateuszbuda/lgg-mri-segmentation

Attention U-Net

https://arxiv.org/abs/1804.03999

EfficientNet

https://arxiv.org/abs/1905.11946

BRISC2025 Dataset

https://www.kaggle.com/datasets/briscdataset/brisc2025